

Lecture 10



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Engineering Mathematics III

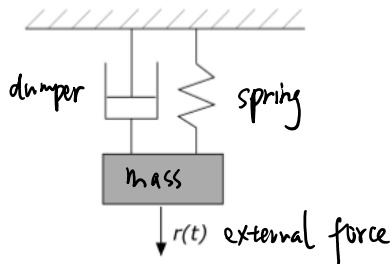
ENGE702

Lecture 10: Second Order Nonhomogeneous ODEs

Second order ODEs

In this lecture we will revisit our mass - spring - damper problem, but in addition to the usual assumptions, there is an external force acting on the mass. The usual form of this external force in practice will be a sinusoid, this is what we will concentrate on, and this is the most interesting case, especially when paired with an underdamped system.

Our system now looks like this:



Denote the displacement from the equilibrium by y .

Second order ODEs

Recall that with no external forces, we have the differential equation

$$my'' + cy' + ky = 0.$$

We got to this by summing the forces on the mass. Now we have an external force to add that (we are assuming) does not depend on y or its derivatives.

We can then add the external force $r(t)$ to the right side of the equation, and we have the second order, nonhomogeneous differential equation with constant coefficients

$$my'' + cy' + ky = r(t).$$

The force from spring is $F_s = -ky$ ^{- spring constant}

The force from damper is $F_d = -cy'$ ^{damper constant}

The external force is $r(t)$

Newton's Law :

$$my'' = -ky - cy' + r(t) \rightarrow \text{combined force}$$

$$\Rightarrow my'' + cy' + ky = r(t)$$

2nd order inhomogeneous linear ODE.

Second order ODEs

The most interesting case is when $r(t)$ is periodic, and we have the equation

$$my'' + cy' + ky = F_0 \cos(\omega t) = v(t)$$

We can find the solution to this as the sum of the corresponding homogeneous equation solution y_h and a particular solution y_p .

The solution y_h will depend on whether the system is not damped, underdamped, overdamped, or neutrally damped.

We can however, find a particular solution y_p .

Second order ODEs

From the table (in the last lecture), we know that our particular solution will be of the form

$$y_p = a \cos(\omega t) + b \sin(\omega t).$$

From here, we can take derivatives of y_p , and put it into our differential equation, and solve for a and b , and we get:

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2} \quad \text{and} \quad b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

where

$$\omega_0 = \sqrt{\frac{k}{m}}$$

is called the 'natural frequency'.

Text Book References

Chapter 2.8

For $my'' + cy' + ky = F_0 \cos(\omega t)$, the right side external force is $v(t) = F_0 \cos(\omega t)$.

so our guess for particular solution is

$$y_p = a \cos(\omega t) + b \sin(\omega t).$$

$$\Rightarrow y_p' = -a\omega \sin(\omega t) + b\omega \cos(\omega t)$$

$$y_p'' = -a\omega^2 \cos(\omega t) - b\omega^2 \sin(\omega t)$$

Substitute back to ODE:

$$-ma\omega^2 \cos(\omega t) - mb\omega^2 \sin(\omega t)$$

$$-caw \sin(\omega t) + cbw \cos(\omega t)$$

$$+ ka \cos(\omega t) + kb \sin(\omega t) = F_0 \cos(\omega t)$$

$$\Rightarrow (-ma\omega^2 + cbw + ka) \cos(\omega t)$$

$$+ (-mb\omega^2 - caw + kb) \sin(\omega t) = F_0 \cos(\omega t) + 0 \cdot \sin(\omega t)$$

$$-mb\omega^2 - caw + kb = 0 \quad (1)$$

$$-ma\omega^2 + cbw + ka = F_0 \quad (2)$$

Solve (1) and (2) for a and b values as

$$a = F_0 \frac{m(\omega_0^2 - \omega^2)}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

$$b = F_0 \frac{\omega c}{m^2(\omega_0^2 - \omega^2)^2 + \omega^2 c^2}$$

$$\omega_0 = \sqrt{\frac{k}{m}}$$

So $y_p = a \cos(\omega t) + b \sin(\omega t)$